



SEA-BIRD
SCIENTIFIC

SBE37-SMP MicroCAT

Instrument Configuration

Instrument Serial Number: 37-16306
Instrument Firmware Version: 4.1
Zero Conductivity Frequency: 2811.19
Communications Format: RS232
Communications Settings: 9600 baud, 8 Data Bits, No Parity

Installed Devices/Sensors

<i>Data Format</i>	<i>Measurement</i>	<i>Sensor Type</i>	<i>Serial Number</i>	<i>Rating</i>
Count	Temperature	Internal	N/A	N/A
Frequency	Conductivity	Internal	N/A	N/A
Count	Pressure Sensor	Druck	10746538	2000m(2000 dBar)

Maximum Depth: **2000m**

CAUTION - The maximum deployment depth will be limited by the measurement range of the pressure sensor, if installed, an attached sensor, if installed, or the housing.



Sea-Bird Scientific
13431 NE 20th Street
Bellevue, WA 98005
USA

+1 425-643-9866
seabird@seabird.com
www.seabird.com

SENSOR SERIAL NUMBER: 16306
CALIBRATION DATE: 27-Oct-17

SBE 37 V2 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

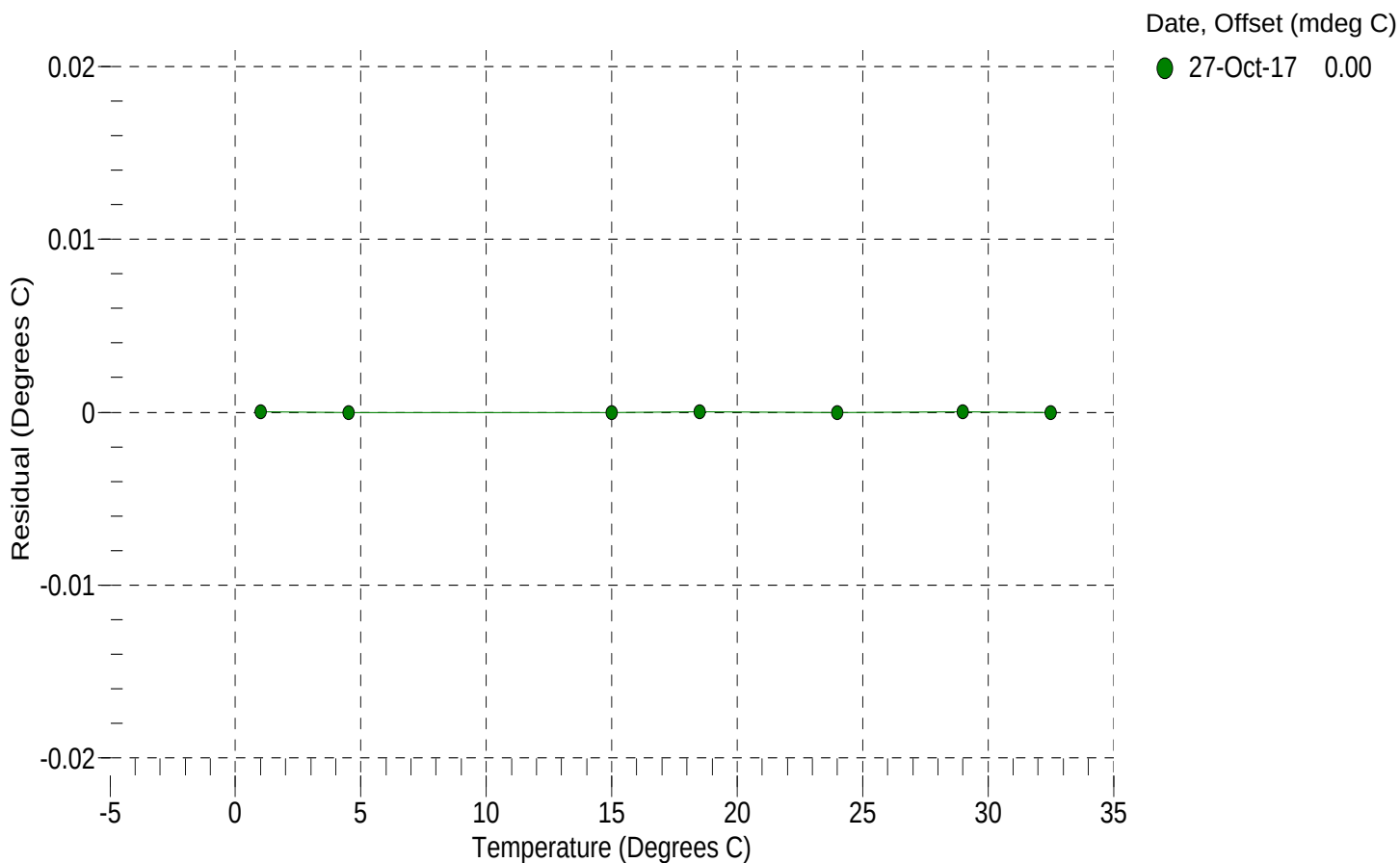
a0 = -1.060390e-004
a1 = 3.050850e-004
a2 = -4.236699e-006
a3 = 1.961165e-007

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	565963.6	1.0000	0.0000
4.4999	484416.2	4.4999	-0.0000
15.0000	309742.2	15.0000	-0.0000
18.5000	268510.8	18.5000	0.0000
23.9940	215855.9	23.9940	-0.0000
29.0000	178004.8	29.0000	0.0000
32.5000	156075.4	32.5000	-0.0000

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1 / \{a_0 + a_1[\ln(n)] + a_2[\ln^2(n)] + a_3[\ln^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature





Sea-Bird Scientific
13431 NE 20th Street
Bellevue, WA 98005
USA

+1 425-643-9866
seabird@seabird.com
www.seabird.com

SENSOR SERIAL NUMBER: 16306
CALIBRATION DATE: 27-Oct-17

SBE 37 V2 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

g = -9.826290e-001
h = 1.245317e-001
i = -1.388808e-004
j = 2.514236e-005

CPcor = -9.5700e-008
CTcor = 3.2500e-006
WBOTC = -1.1856e-007

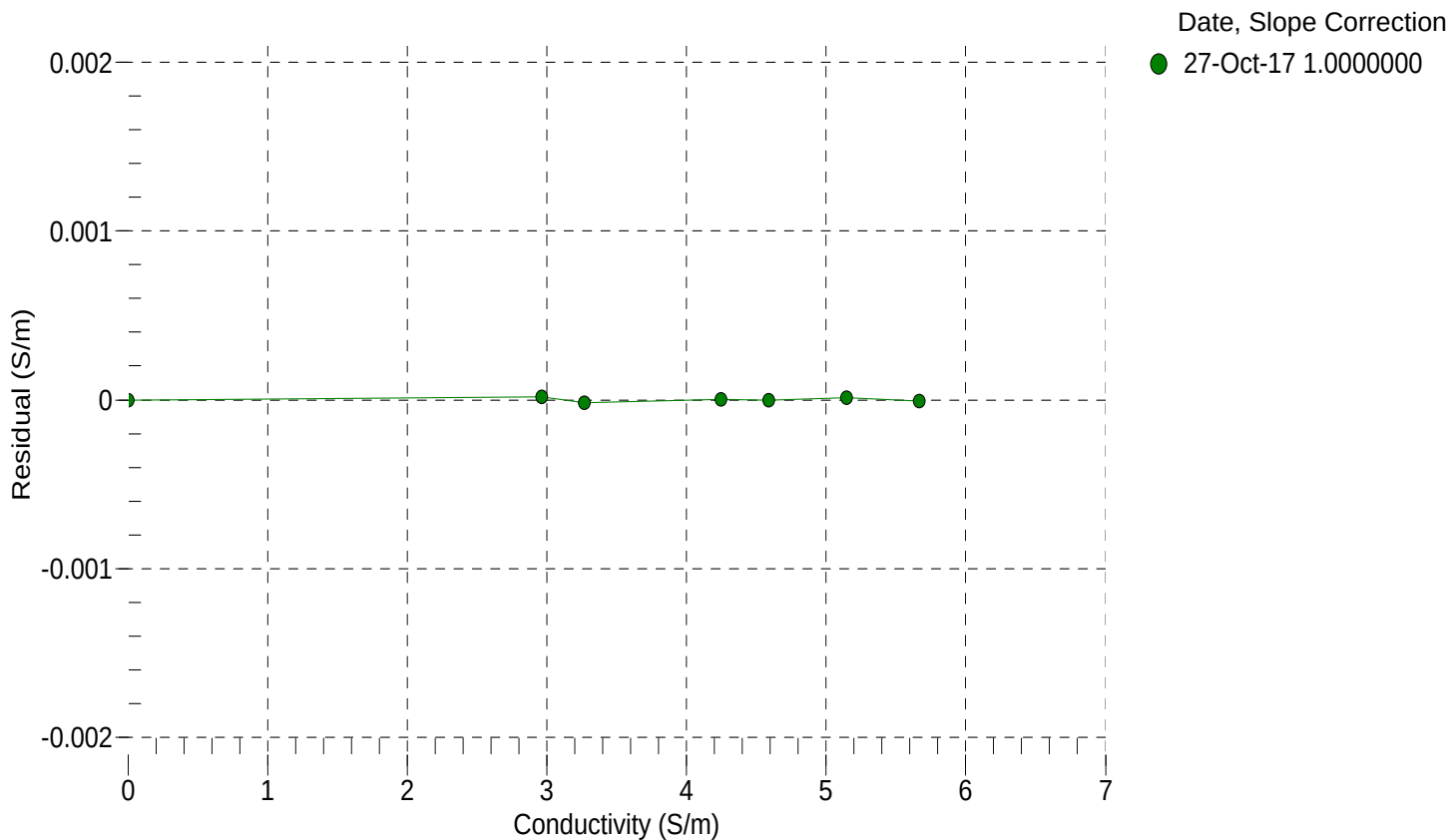
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2811.19	0.00000	0.00000
1.0000	34.6440	2.96264	5628.26	2.96265	0.00002
4.4999	34.6248	3.26842	5841.56	3.26840	-0.00002
15.0000	34.5827	4.24598	6475.78	4.24598	0.00000
18.5000	34.5741	4.58969	6684.20	4.58969	-0.00000
23.9940	34.5647	5.14470	7007.38	5.14472	0.00001
29.0000	34.5594	5.66496	7296.98	5.66495	-0.00001
32.5000	34.5552	6.03559	7496.22	6.03545	-0.00014

$f = \text{Instrument Output(Hz)} * \sqrt{1.0 + \text{WBOTC} * t} / 1000.0$

t = temperature (°C); p = pressure (decibars); $\delta = \text{CTcor}$; $\epsilon = \text{CPcor}$;

Conductivity (S/m) = $(g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \epsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity





Sea-Bird Scientific
13431 NE 20th Street
Bellevue, WA 98005
USA

+1 425-643-9866
seabird@seabird.com
www.seabird.com

SENSOR SERIAL NUMBER: 16306
CALIBRATION DATE: 19-Oct-17

SBE 37 V2 PRESSURE CALIBRATION DATA
2900 psia S/N 10746538

COEFFICIENTS:

PA0 =	-4.428071e-002	PTCA0 =	5.241446e+005
PA1 =	8.899179e-003	PTCA1 =	-2.351636e+000
PA2 =	-1.529360e-010	PTCA2 =	9.330739e-002
PTEMPA0 =	-7.977824e+001	PTCB0 =	2.523775e+001
PTEMPA1 =	4.558922e-002	PTCB1 =	3.500000e-004
PTEMPA2 =	-1.523858e-008	PTCB2 =	0.000000e+000

PRESSURE SPAN CALIBRATION

THERMAL CORRECTION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)	TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
14.40	525766.6	2263.6	14.42	0.00	32.50	2465	525908.14
591.65	590723.1	2266.3	591.61	-0.00	29.00	2388	525897.85
1168.79	655818.5	2267.5	1168.75	-0.00	23.99	2278	525884.35
1745.94	721065.3	2268.2	1745.92	-0.00	18.50	2157	525874.95
2323.01	786452.7	2269.1	2323.03	0.00	15.00	2080	525870.82
2900.11	851979.0	2270.1	2900.06	-0.00	4.50	1850	525880.40
2323.02	786456.5	2269.4	2323.07	0.00	1.00	1773	525882.92
1745.94	721072.1	2268.7	1745.98	0.00	TEMPERATURE (°C) SPAN (mV)		
1168.85	655831.4	2268.6	1168.86	0.00			
591.70	590730.4	2268.3	591.68	-0.00			
14.40	525766.6	2269.0	14.42	0.00			

y = thermistor output (counts)

t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y²

x = instrument output - PTCA0 - PTCA1 * t - PTCA2 * t²

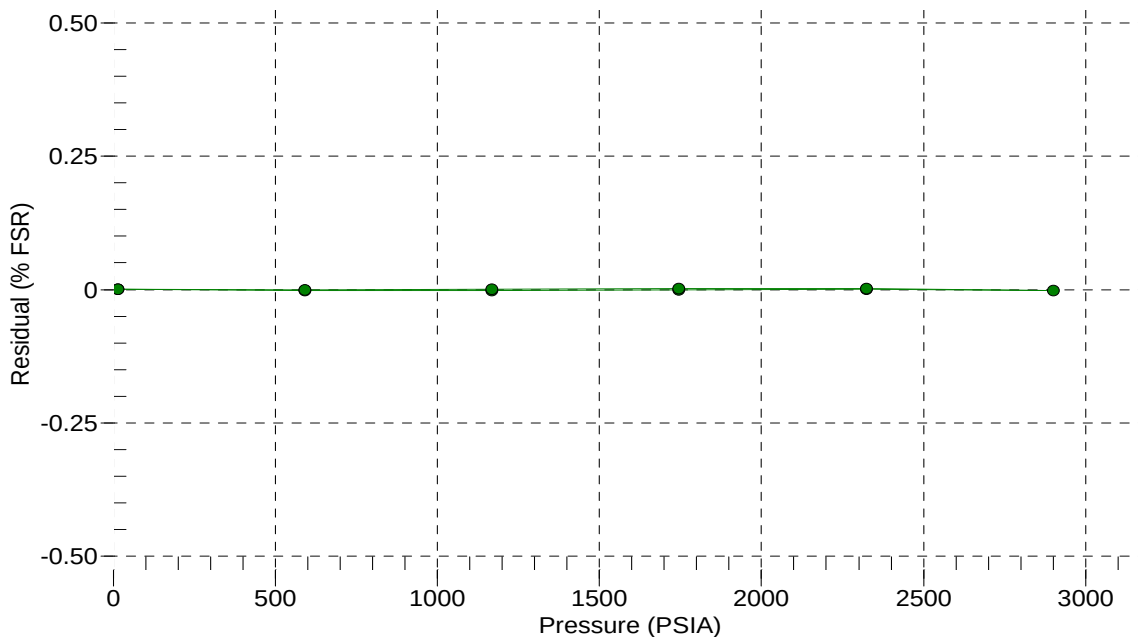
n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t²)

pressure (PSIA) = PA0 + PA1 * n + PA2 * n²

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range

Date, Offset (%FSR)

● 19-Oct-17 0.00





Sea-Bird Scientific
13431 NE 20th Street
Bellevue, WA 98005
USA

+1 425-643-9866
seabird@seabird.com
www.seabird.com

Pressure Test Certificate

Test Date: **2017-10-18**

Description: **SBE-37 Microcat**

Sensor Information:

Model Number: **SBE-37**

Serial Number: **16306**

Pressure Test Protocol:

Low Pressure Test: **40** PSI Held For: **15** Minutes

High Pressure Test: **2900** PSI Held For: **15** Minutes

Passed Test: **True**

Tested By: **JY**

